H3. Predicting the 30-Day Readmission — And Doing Something About It

Real-world pain: A meaningful share of discharged patients bounce back within 30 days. Many of those returns were preventable with a phone call, a medication check, or a follow-up appointment. But a hospital's care-coordination team can only reach a handful of patients a day.

The problem: Predict 30-day readmission risk at the moment of discharge, and — more importantly — turn that risk into a ranked, resource-constrained intervention list.

What we provide:

- 25k synthetic discharge records: demographics, primary and secondary diagnoses (coded), length of stay, prior admissions, procedures, discharge disposition, medication count, key lab values
- Deliberate real-world messiness: ~15% missing labs, inconsistent coding, some duplicate patient records
- Hidden 30-day readmission outcome
- A stated constraint: the care team can call **only 50 patients per week**

Assumptions students may make:

- Post-discharge behaviour (whether they filled the prescription, attended follow-up) is unobserved
- Socioeconomic and distance-from-hospital data is unavailable
- Some readmissions are planned/unavoidable — teams should reason about whether these should be excluded
- Missingness may itself be informative (a lab not ordered tells you something)

Desired output:

1. A calibrated risk score per patient — **calibration is explicitly scored**, because a "70%" that means 30% is useless for prioritisation
2. `evaluate.py` score = AUC-PR plus **recall@50** (the metric that respects the real constraint), plus a calibration error term
3. Per-patient drivers of risk, expressed as something a nurse can act on ("11 medications, no follow-up scheduled, third admission in 6 months")
4. A **bias audit across age, sex, and payer type** — required, not optional. Report performance gaps and describe mitigation
5. A one-page "Monday morning call list" for the care team, and a rough estimate of readmissions averted

Stretch: recommend a specific intervention type per patient (pharmacist call vs. home visit vs. appointment booking) and justify the mapping.